

An Intro to Big Data and Machine Learning for Survey Researchers and Social Scientists

Trent D. Buskirk, Ph.D. (Marketing Systems Group)

Frauke Kreuter, Ph.D. (University of Maryland)

SVY 751, Notes Part 1



Analyses of Big Data

Classical Statistical Approaches versus Statistical Machine Learning

Model Evaluation/Validation

Popular machine learning techniques

- K-Means Clustering

- K Nearest Neighbors

- CARTS

- Random Forests

Analysis in a BIG DATA World

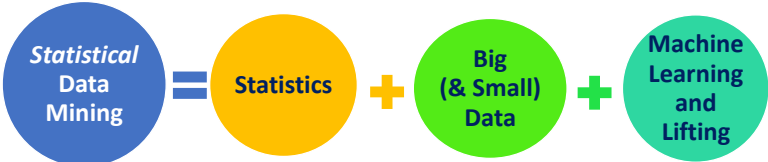
- Many classical statistical techniques you learned while in Grad School, on the job, or from conferences, courses and in life are aptly suited for smaller and moderately sized data
- In the big data paradigm, however, some of these well relied upon techniques may not
 - Differentiate between significant and non-significant predictors (i.e. every variable, whether meaningful or spurious could be retained in your final models)
 - Every correlation measure is significantly different from zero
 - Provide visually informative displays between relationships (i.e. the over crowded scatterplot)

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Big Data

Data Mining Paradigm

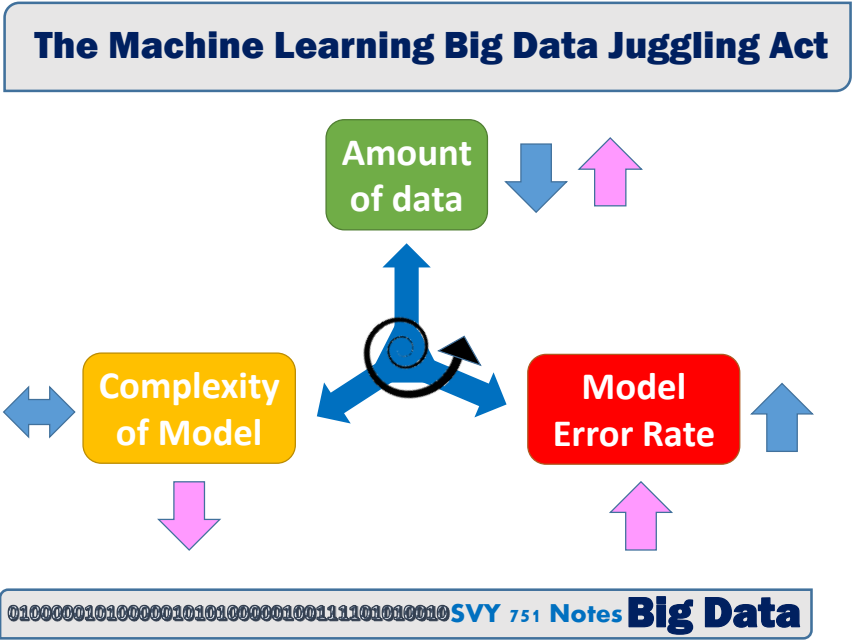
- Data Mining is a process that combines
 - Statistical Analysis (Exploratory Data Analysis) with
 - Big (and Small) Data with
 - Machine Learning and lifting



- Lifting focuses on the rows of the data table/set (i.e. the respondents or units that are observed)
- Learning focuses on the columns of the data set (i.e. those variables that were observed explicitly (e.g. survey questions) or implicitly (e.g. paradata, appended data) and leverages computational algorithms to find structures among these variables.

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Machine Learning/Data Mining versus Statistics

Methods Aspect	Data Mining/Machine Learning	Statistics
"Large Data"	1,000,000 rows, 4000 columns	1,000 – 5,000 rows, 30-500 columns
Data Collection	Administrative, Search Queries, Third Party, Convenient	Surveys, Designed Experiments, Observational Studies, Panels
Sample Data?	Sometimes – but "estimates" can be expedited through parallel computing power/cloud computing	Yes – frameworks provide mechanism for computing standard errors

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Machine Learning/Data Mining versus Statistics

Methods Aspect	Data Mining/Machine Learning	Statistics
Modelling	Flexible – few, if any assumptions on underlying distribution	Particular model and error structure specified @ analysis (NP vs. P)
Inference	Prediction/Classification usually main focus	Understanding/explaining/refining underlying theory; confidence intervals, parameter estimates
Computational Aspects	Computational efficiency important and considered	Computational efficiency not usually primary focus
Variable Selection/Model fitting	Variable selection and overfitting models can be problematic	Variable and model selection issues must be considered

Machine Learning Overview

- **Two Types of “Learning”**
 - **Supervised Learning –**
 - Dataset has clear set of Predictors (Xs) and Outcomes (Y)
 - Goal is to understand the relationship between predictors and outcomes
 - End goal is either Inference or Prediction
 - **Unsupervised Learning –**
 - Dataset consists of Predictors (Xs) only
 - Think of Sampling Frame Data – all auxiliary variables – no survey outcome variables (i.e. Ys)
 - We seek to understand the relationships between the variables/observations in the Dataset

Supervised Learning and Survey Data

- **Smartphone surveys have begun capturing photo data from respondents (Michaud, Buskirk and Saunders and Nielsen, 2012)**
- **Suppose a survey asked respondents about their “snacking habits” and asked them to snap a photo of the snack they are currently working on**
 - ▢ **Snacks could be classified as edible or drinkable.**
 - ▢ **Coders could mark photos as edible or drinkable and then the photo data captured from surveys could be classified into one of these categories using the machine learning model that was generated from a training/pilot set of images.**

Stanford team creates computer vision algorithm that can describe photos (2014)
<http://news.stanford.edu/news/2014/november/computer-vision-algorithm-111814.html>

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Unsupervised Learning and Survey Data: Text Analysis

- **Twitter data can be mined from the web using various resources including Python functions and R functions such as TwitterR.**
- **Text analytics apply unsupervised learning algorithms to group like tweets**
 - ▢ **“similar tweets” could be length, location, content, time, and shared followers, for example.**
 - **Natural Language Processing**
 - ▢ **sentiment analysis is accomplished by coding a set of tweets and training a machine learning classifier (supervised learning).**
 - **The trained classifier is then applied to future or new tweets and sentiment is ascribed to these as “predictions” from the trained model.**

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Unsupervised Learning Example

- **Survey Researchers can use para-data obtained from a survey panel (probability or non-probability) to understand whether there are groups of panelists that are similar on certain survey attributes including:**
 - **Survey length**
 - **Incentive amount**
 - **Field period length**
 - **Number of Invitations required before response**
 - **Other attributes**
- **The idea would be to determine if there is a group of panelists who consistently take longer surveys, for more money versus those who take shorter surveys for less money, for example.**

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Some Ideas for using Machine Learning in Survey Research

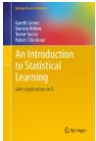
- **Imagine wanting to sample villages within a country for which little infrastructure or previous info is available.**
 - You can obtain Route Information from a local courier service there. Generally, the data contain volume of deliveries, frequencies and length of stopover.
 - **For a greater fee, you obtain additional information on the type of delivery point for half the records (village, business, school, hospital).**
 - Through machine learning algorithms you develop a classification model that groups delivery points into corresponding Structure Types: Villages or otherwise.
 - **Based on this model, you can determine which of the remaining delivery points are also villages.**
 - The final collection of delivery points identified as villages becomes your sampling frame.

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Data Mining/Machine Learning Resources

- <http://www.dataminingconsultant.com/resources.htm>
- **Data Mining Algorithms Explained Using R (2015)**
 📖 <http://bit.ly/1yZYHjK>
- **Data Mining for the Social Sciences (2015)**
 📖 <http://bit.ly/1DpPFC2>
- **An Introduction to Statistical Learning with Applications in R (2013)**
 📖 **Free PDF Version:** <http://bit.ly/1iUJso0>
 📖 **Online Resources for FREE lecture videos and labs in R**
 ◦ <http://bit.ly/1snBMk5>
- **An overview of Machine Learning Functions available in R**
 📖 <http://cran.r-project.org/web/views/MachineLearning.html>



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Analyses of Big Data

- Classical Statistical Approaches versus Statistical Machine Learning
- Model Evaluation/Validation
- Popular machine learning techniques
 - K-Means Clustering
 - K Nearest Neighbors
 - CARTS
 - Random Forests

Prediction versus Explanation

- Many applications of classical statistics have focused on explaining how or possibly why certain predictors are related to outcomes we care about
 - Descriptive statistics quantify the degree of association or relationship.
 - Inferential statistics provide a way to confirm that the relationship goes beyond random error.
 - We evaluate models based on their *explanatory power* (R^2).
- Machine learning algorithms can also describe relationships between predictors and outcomes
 - But many times, this is not the end goal.
 - End goal is to develop well tuned predictive “models” that can be applied to new data.
 - Here models are evaluated based on their *predictive power*.
- An excellent overview of the model building process within the context of prediction is given by Shmueli (2010).
 - See: <http://bit.ly/1F27hGT>

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Building and Evaluating Models with Machine Learning Algorithms in the Big Data Context

- Step 1: Determine the Appropriate Model/ML Method for the task at hand/outcome of interest
 - Beware of the dreaded TYPE III error
 - Providing the right answer to the wrong question or “having correctly solved the wrong problem”
- Step 2: Determine strategy for training/tuning/calibrating and evaluating models.
- Step 3: Estimate model parameters (i.e. train models)
- Step 4: Estimate other model “tuning” parameters that can’t be directly inferred or computed from the data at hand
- Step 5: Evaluate and use the model (apparent performance, internal/ temporal/ external validity)
 - This step will be related to step 2 above!

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Steps 2/5: Determining a Calibration/Tuning and Evaluation Strategy for ML Models

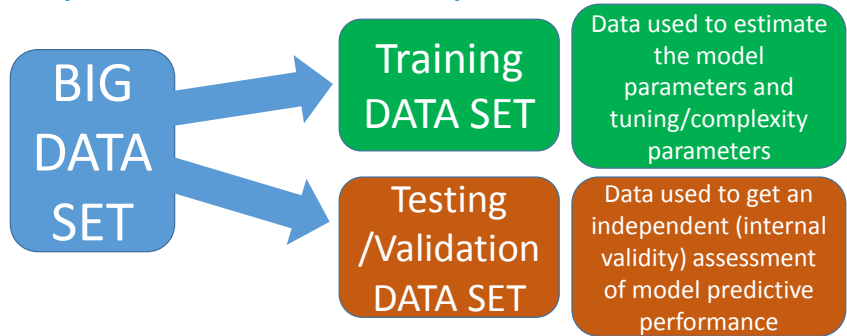
- In general, cross-validation methods create resamples of the data
- The machine learning method or model is “trained” on one set and then applied to or tested on the other.
- Much like replication methods used for understanding the variance of estimators in the survey framework, cross-validation methods seek to understand model predictive performance by sub-sampling from the original data.

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Steps 2/5: Determining a Calibration/Tuning and Evaluation Strategy for ML Models, cont.

- The most simple method for model validation is a simple split sample (training/testing)
 - Training sample should be large enough to handle complexity of the model, but not too large to eclipse the overall distributions of predictor variables in the test sample (Molinaro et al., 2005).



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Steps 2/5: Determining a Calibration/Tuning and Evaluation Strategy for ML Models, cont.

- A very common and effective alternative to the split sample validation method used in the machine learning literature and beyond is K-fold cross-validation. (Common value of K is 10).
- Cross-validation has been shown to be superior to simple split sample approaches for model validation
 - ▢ (See Kuhn and Johnson, 2013, pg. 67 for more references).

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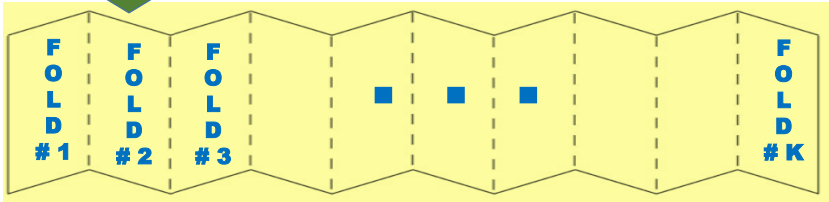
Big Data

K-Fold Cross Validation

BIG DATA SET

Create
K Folds

A fold is a subset selected at random without replacement from the big data set. In the case of 10-fold validation, a fold represents a 10% subsample.



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Working with K-Fold Cross Validation

- If you have ever worked with Delete-A-Group jackknife then the evaluation of performance using K-fold cross validation will be very intuitive.
- K-fold cross validation method proceeds iteratively:
 - the first fold becomes the test sample and the remaining 9 folds are combined to form the training sample. The model/method is fit/applied to the training sample and performance is evaluated on the test sample.
 - the process continues using each fold as the test sample and the remaining K-1 folds as the training sample
 - the K values of model performance obtained from each fold used as the testing set are then averaged together to form an estimate of cross-validated performance. Variability in these performance metrics can also be obtained using the K-values.

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Predictive Validity within the Big Data Context

- So why do we care about split sample or other validation methods when working with big data?
 - Large sample sizes generally result in most predictors being statistically significant (possible Type I errors)
 - If the models built using the training sample were evaluated for fit using the training sample itself, the overall error rates would be overly optimistic (Kuhn and Johnson, 2013).
 - This type of validity or evaluation is referred to as apparent performance or apparent validity.
 - See Steyerberg and Harrell: <http://bit.ly/1K7IIWJ>

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Predictive Validity in the Survey Research Context

- Within the context of survey research, we should think more critically about the level of predictive validity we require for various survey related tasks.
 - For weighting purposes, internally valid models are most needed – ones that are consistent and accurate for a given sample of respondents and nonrespondents.
 - For tailored and responsive survey designs, predictive models should be internally, temporally, and in some cases, externally valid because their application is intended at some subsequent point in time or possibly to a subsequent sample.
 - Prediction of Respondents to future survey based on survey respondents historical data (panel) or paradata
 - Buskirk, Burks and West, 2013;
 - Wagner and Frost, 2014;
 - Schouten et al., 2013

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Unsupervised Learning in the Survey Context

- Generally, a mix of both unsupervised and supervised machine learning approaches can be used for a given study:
 - Auxiliary information could be used to form clusters that were optimal (most heterogeneous) and then survey results could be obtained by incorporating the sampling design (Elliot, 2011: <http://1.usa.gov/1OUlyJd>)
 - Tailored survey design to predict survey response as a function of hundreds of geographic auxiliary data appended to an ABS sample – use Principal Components to “reduce” the variables into a smaller set of independent signals to be used in a logistic regression model (See Buskirk, West and Burks, 2013).
 - Using geographic variables to predict presence of phone number for a sampling frame of addresses – Olson and Buskirk (2015) first applied factor analysis to create smaller set of factors used for prediction (See Olson and Buskirk, 2015).

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http://bits.blogs.nytimes.com/2015/03/17/ibm-introduces-twitter-fueled-data-services-for-business/?_h=21194ac2ead6079e9e00176c20937f334d92ea63&ts=1426600932&r=0

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-
- Per capita consumption of mozzarella cheese (US)
- Civil engineering doctorates awarded (US)
- Correlation: 96% Sources: USDA & National Science Foundation tylerjvigen.com
- | Year | Per capita consumption of mozzarella cheese (US) (Pounds) | Civil engineering doctorates awarded (US) (Degrees awarded) |
|------|---|---|
| 2000 | 9.3 | 480 |
| 2001 | 9.7 | 500 |
| 2002 | 9.7 | 520 |
| 2003 | 9.7 | 530 |
| 2004 | 9.9 | 540 |
| 2005 | 10.2 | 600 |
| 2006 | 10.5 | 640 |
| 2007 | 11.0 | 680 |
| 2008 | 10.6 | 700 |
| 2009 | 10.6 | 700 |

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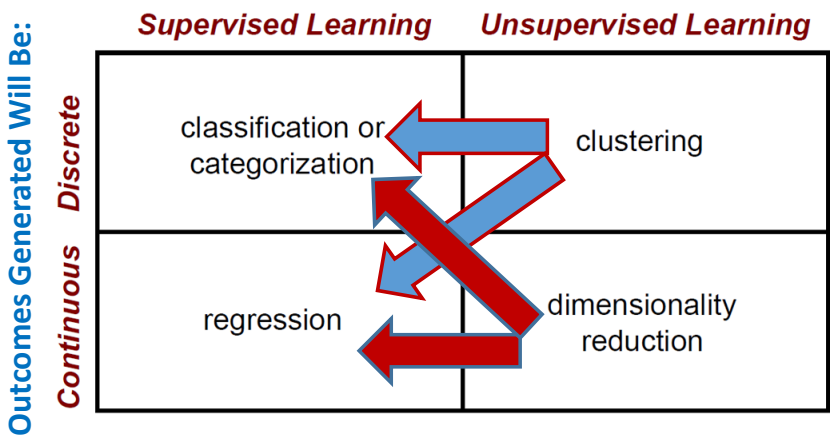
CARTS

Random Forests

Machine Learning Methods/Techniques

- There are many different machine learning methods available
- Many are non-parametric in nature and while a functional form can be specified, it is generally not a natural byproduct of the method
- Wu et al. (2008) provide an overview of ten of the top machine learning algorithms including (see <http://bit.ly/1liWTir>) :
 - ★  K-means Clustering
 - ★  PageRank
 - ★  K-nearest neighbors
 - ★  Support Vector Machines
 - ★  Decision Trees and Classification and Regression Trees
 - ★  Apriori Algorithm
 - ★  The EM Algorithm (Expectation-Maximization)
 - ★  Naïve Bayes
 - ★  Ensemble Methods (like AdaBoost and Random Forests).

Overview of Machine Learning Methods/Outcomes



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A Basic Example Applying Machine Learning to Sample Designs

- Imagine that the MobileMaddness Phone Company wanted to determine the best way to communicate with its customers who have enrolled in its VIP program.
- The company believes social network micro messages will be more salient than other avenues. It wants to estimate the average amount of time spent on social networking sites via mobile apps and web for its VIP customers.



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**A Basic Example Applying Machine Learning
to Sample Designs, Cont.**

- A data file for VIP members contains email address, past month mobile internet activity, data consumption in past month, text message volume and monthly bill.
- The company believes that the amount of time spent on social networking sites (on mobile devices) may be related to overall mobile internet activity, data consumption and perhaps total bill.
- The data file contains millions of records with information available for the 5 variables mentioned.
- Goal: Efficiently sample records to send a two question survey about total time spent last month using (a) social networking mobile sites and (b) apps.

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**A Basic Example Applying Machine Learning
to Sample Designs, Cont.**

- One possible solution:
 - Rather than take a simple random sample of X% of the records, it could be more efficient to first stratify the total VIP Customer frame.
 - But the auxiliary variables that are likely related to the key outcome of interest (time on social media mobile sources) are continuous.
 - We could create categorical versions of each of the three key predictors then cross them to form exhaustive strata
 - A possibly more efficient way could be to cluster the VIP customers using the three variables: total bill, total data consumed and mobile internet activity.
 - The resulting “clusters” become the strata for sampling.

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Big Data

Cluster Analysis – Not Cluster Sampling

● **Two main types of cluster analysis:**

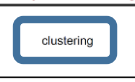
■ **K-means Clustering**

- Seek to partition observations into a pre-specified number of clusters (groupings)
- Number of clusters is known *a priori* (i.e. parameter of the method)

■ **Hierarchical Clustering**

- The number of clusters is not known in advance
- By product of this method is a Dendrogram – a tree like plot that illustrates the formation of all resulting clusters.

Unsupervised Learning



Cluster Observations Using their Features (Variables) → Subgroups among the Observations (Segments)

Cluster Features (Variables) Using all observations → Subgroups among the Features (Profiles)

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Method Overview: K-means Clustering

● **What type of Learner?**

- Nonparametric method; Unsupervised Learner
- Method is also known as “data segmentation”



● **For what problems can you apply this method?**

- Aims at finding discrete categories/groupings based on a collection of covariates (no outcome variables used) that are measured on a continuous scale (or at least ordinal).

● **Does this method have any Tuning Parameters?**

- One parameter: k – the number of CLUSTERS (known in advance, or at least a range of clusters).
- And the distance measure used to determine proximity/similarity between cases.

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Method Overview: K-means Clustering

● How does this Method Work?



INITIALIZATION Step: Each case is randomly assigned to one of K clusters (K is known and provided by the user)

ITERATION Step 1: Computation – For each of the K clusters, compute the Cluster Centroid (component-wise means)

ITERATION Step 2: Reclassification – Assign each case to the cluster whose centroid is the closest as measured by squared Euclidean distance (or other distance function).

FINALIZATION Step: The algorithm “converges” when cluster reclassification stop changing

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Method Overview: K-means Clustering

● Insights on Parameters for this Method



- 📖 Small k → possibly more heterogeneous (and larger) clusters;
- 📖 Large k → smaller clusters, more homogeneous with respect to variables used in determining clusters;
- 📖 The squared Euclidean distance is by far the most common distance measure. Specifying the appropriate distance/dissimilarity measure is by far more important than the choice of clustering algorithm.
- 📖 Technically, the initial cluster assignments can be a third parameter. Usually, the algorithm is run using different starting seeds (i.e. different initial cluster assignments) for the same value of K. Results are compared to assess stability of the solution/sensitivity to starting assignments.

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Method Overview: K-means Clustering

- **Caveats/ Considerations for the Method**
- 
- **This method can be sensitive to the scaling of the p-predictors. If all are on different scales/ measured in different units, standardizing variables is recommended.**
- **This effectively gives every predictor equal weight in determining “similarity” of cases.**
- **This method is not recommended for use with nominal categorical predictors without some additional preparation/transformation of these variables.**
- **One alternative is k-mode clustering or providing similarity matrix for each case.**

Method Overview: K-means Clustering

- **Implementation of this method in R**
- 
- `kmeans {stats package};`
- `cluster.Sim {clusterSim package};`
- `knn.reg {cluster package}`
- `rattle {rattle package}`
- <http://www.r-bloggers.com/k-means-clustering-from-r-in-action/>
- **Examples of this method in the S.R. Literature**
- **Segmentation of Clients and Client Interests using Survey Data**
- **Krantz et al. (2009);** <http://bit.ly/1OIrnAd>
- **Developing a typology of Internet Use based on Survey Data**
- **Bae Brandtzaeg; et al. (2011);** <http://bit.ly/1FkyhQr>
- **Frame improvements—identifying missing subtypes of Farms**
- **McCarthy et al. (2009);** <http://1.usa.gov/1lcQekK>

Cluster Analysis Example #1

● A new poll is desired for the state of Florida regarding income and employment rate disparities in the state’s workforce. The client has specified that comparisons between and distinct estimates from three primary racial/ethnic groups are desired.

- ▢ Groups include: Hispanic, Non-Hispanic Black and Non-Hispanic Other (including White, Asian and other races).
- ▢ The design is a stratified cluster sample in which counties will be stratified into groups and selected. A random sample of addresses within each selected county will occur at the final sampling stage.
- ▢ Auxiliary data for each of the 67 Florida counties include:
 - Percentage of Hispanic Population
 - Percentage of Non-Hispanic Black Population
 - Percentage of Non-Hispanic Other Population



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Cluster Analysis Example, Cont.

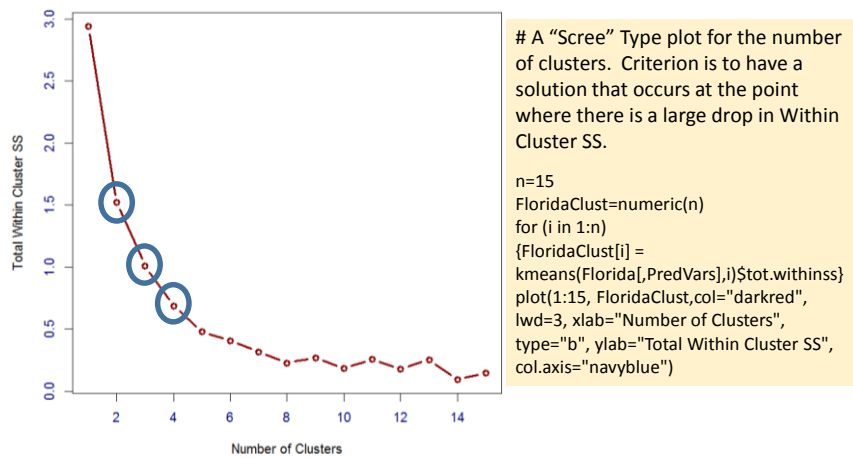
● Task: Can we use the auxiliary data to form strata that will provide sufficient strata to represent the three race/ethnic targets?

- ▢ It would be hard to stratify directly based on race since the sampling occurs in multiple stages
- ▢ Certainly, we could append race data for every address in Florida and then stratify,
 - but this information is not readily available or cost-feasible
- ▢ We could use the auxiliary data to create strata based on cross-tabulating binary “High/Low” race group percentages from each Race/Ethnic Group (or quartiles, etc),
 - but this process may result in too many or sparse strata and may be inefficient as it essentially considers each variable separately.

Elliott, M. R. (2011). A Simple Method to Generate Equal-Sized Homogenous Strata or Clusters for Population-Based Sampling. *Annals of Epidemiology*,21(4), 290–296.
doi:10.1016/j.annepidem.2010.11.016

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A Heuristic for Determining the Number of Clusters

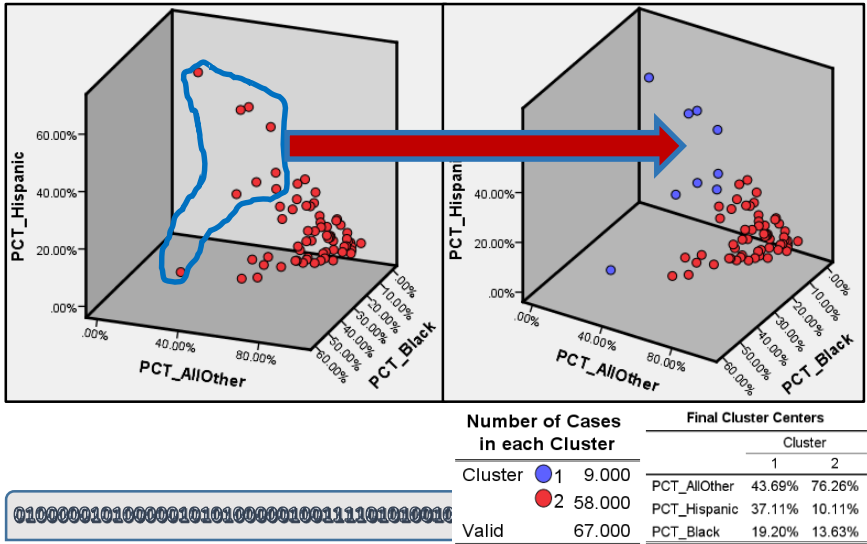


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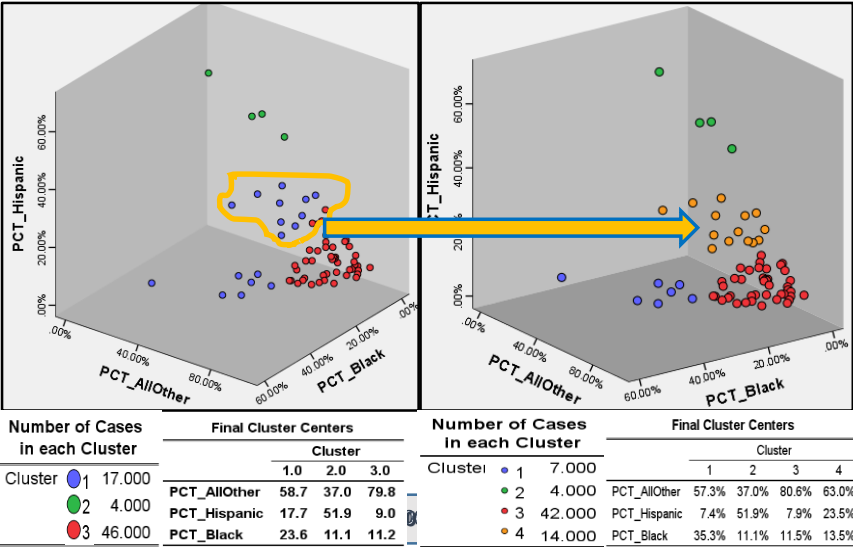
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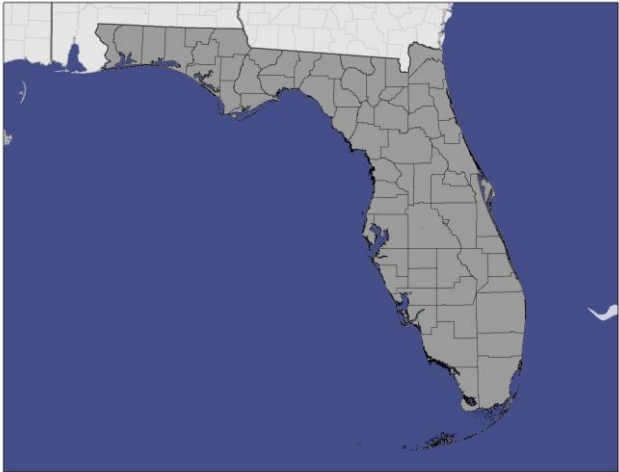
Data and a 2-Cluster Solution



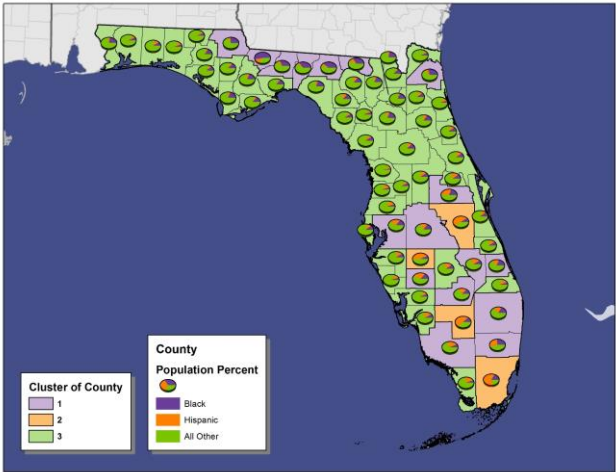
Three and Four Cluster Solution



Visualizing the Clusters



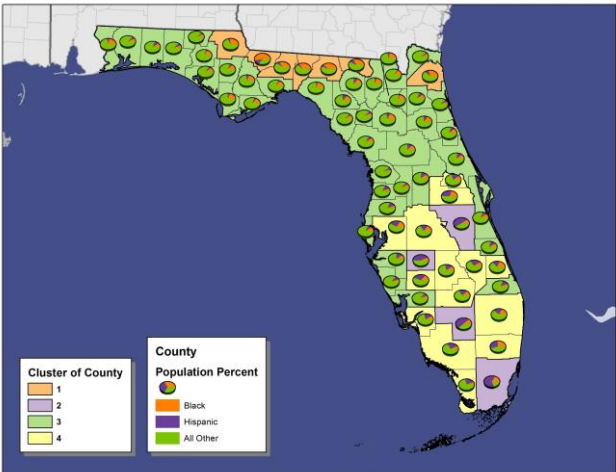
Visualizing the Clusters



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Visualizing the Clusters



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SPSS Syntax for K-means Clustering

```

DATASET ACTIVATE FloridaData.
QUICK CLUSTER PCT_AllOther PCT_Hispanic PCT_Black
/MISSING=LISTWISE
/CRITERIA=CLUSTER(2) MXITER(20) CONVERGE(0)
/METHOD=KMEANS(NOUPDATE)

/SAVE CLUSTER DISTANCE
/PRINT INITIAL ANOVA CLUSTER DISTAN.

GRAPH
/SCATTERPLOT(XYZ)=PCT_AllOther WITH PCT_Hispanic WITH PCT_Black BY QCL
/MISSING=LISTWISE.
```

Number of Clusters
(User Specified)

Requests that a categorical
Variable be created that
identifies the clusters (QCL is
default name)

SPSS Generated
Cluster Variable

R-Code for producing k-means clustering

```

# Cluster Example
# Identify Variables from the Florida Data set that need to be clustered
PredVars<-names(Florida)[c(15,16,19)]
```

```

set.seed(517)
Florida3cluster<-
kmeans(na.omit(Florida[,PredVars]), centers=3)

#Print Number of Cases per Cluster
table(Florida3cluster$cluster)
# 1 2 3
# 4 17 46

#Print Cluster Centers
t(round(100*Florida3cluster$centers,1))
#      1  2  3
#PCT_Hispanic 51.9 17.7 9.0
#PCT_Black    11.1 23.6 11.2
#PCT_AllOther 37.0 58.7 79.8
```

3-Cluster
Solution

```

set.seed(1)
Florida4cluster<-
kmeans(na.omit(Florida[,PredVars]), centers=4)

#Print Number of Cases per Cluster
table(Florida4cluster$cluster)
# 1 2 3 4
#42 4 7 14

#Print Cluster Centers
t(round(100*Florida4cluster$centers,1))
#      1  2  3  4
#PCT_Hispanic 7.9 51.9 7.4 23.5
#PCT_Black    11.5 11.1 35.3 13.5
#PCT_AllOther 80.6 37.0 57.3 63.0
```

4-Cluster
Solution

Cluster Analysis Example 2:

- Of interest is to estimate the number of Facebook Users in the U.S. and in general, which states are more digitally social
 - ▢ Estimate the number of social media users there are in the U.S. (e.g. facebook, pinterest)
- Auxiliary data are limited. One possible source of auxiliary information pertains to search volumes for each state on specific key words or websites:
 - ▢ Facebook search term
 - ▢ Pinterest search term
 - ▢ Twitter search term
 - ▢ Other related search term
- Using historical data on Facebook Users estimated by state, we could determine if the internet search volume data are related to one of our key outcome measures.
- If so, stratification of states on this and related searches might lead to more efficient estimates and sampling designs.

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Big Data

Cluster Example 2, Cont.

- According to Pew, 2014 Social Media Update, of Online Adults:
 - ▢ 71% use Facebook
 - ▢ 23% use Twitter
 - ▢ 26% use Instagram
 - ▢ 28% use Pinterest
 - ▢ 28% use LinkedIn

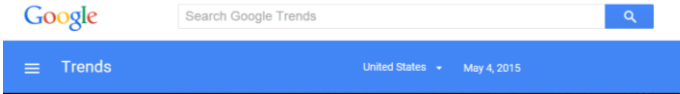
Search Parameters:

Search =Google Trends

Search Term=Facebook,

Search Term2=Pinterest;

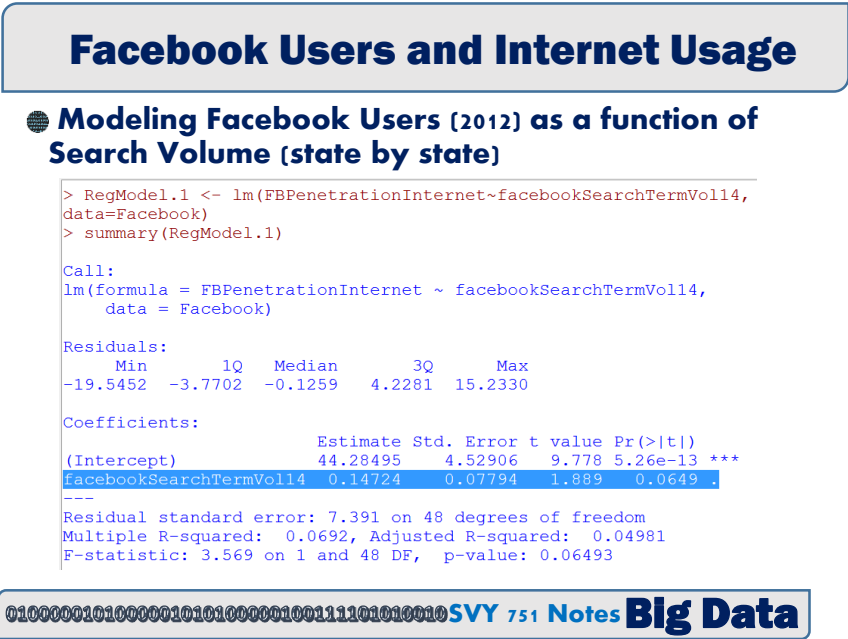
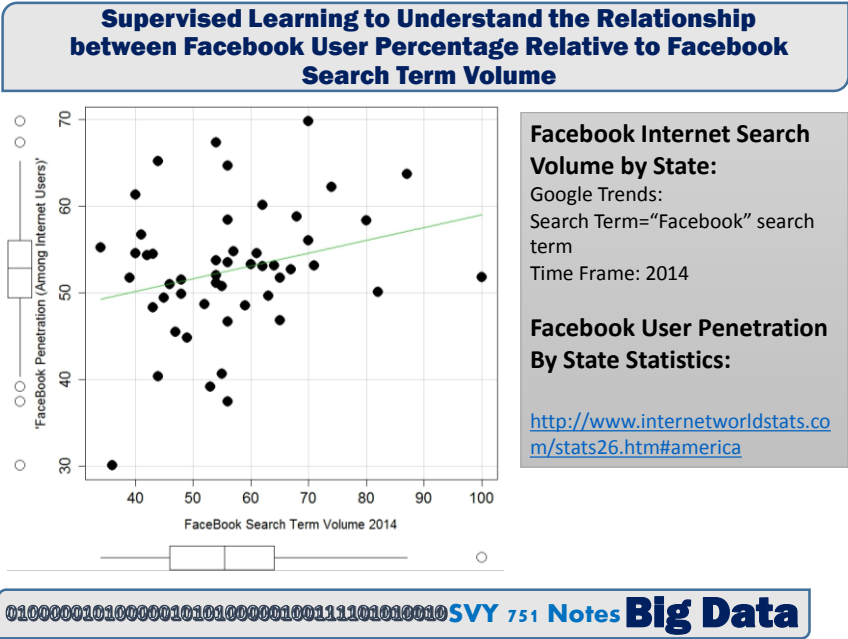
Time Range: 2014



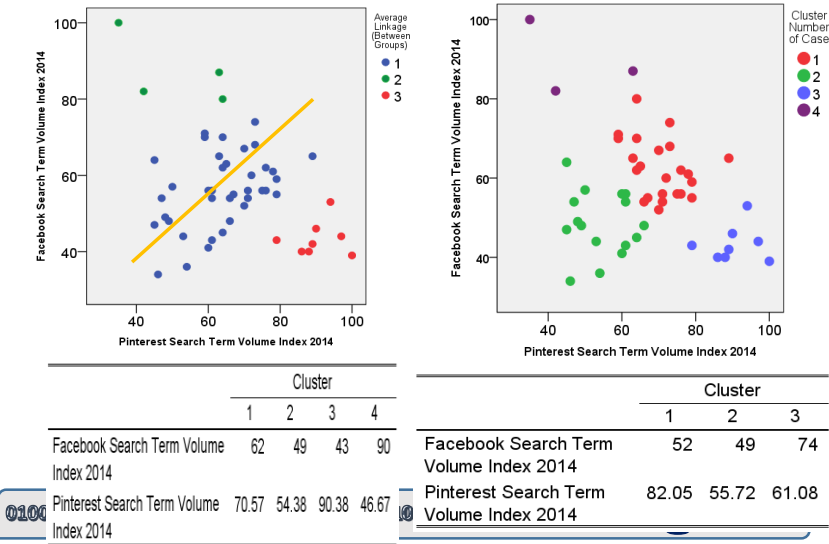
<http://www.pewinternet.org/2015/01/09/social-media-update-2014/>

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Cluster Example 2 cont. :
Social Media Searches



A Basic Example Applying Machine Learning to Sample Designs

- Imagine that the MobileMaddness Phone Company wanted to determine the best way to communicate with its customers who have enrolled in its VIP program.
- The company believes social network micro messages will be more salient than other avenues. It wants to estimate the average amount of time spent on social networking sites via mobile apps and web for its VIP customers.



**A Basic Example Applying Machine Learning
to Sample Designs, Cont.**

- A data file for VIP members contains email address, past month mobile internet activity, data consumption in past month, text message volume and monthly bill.
- The company believes that the amount of time spent on social networking sites (on mobile devices) may be related to overall mobile internet activity, data consumption and perhaps total bill.
- The data file contains millions of records with information available for the 5 variables mentioned.
- Goal: Efficiently sample records to send a two question survey about total time spent last month using (a) social networking mobile sites and (b) apps.

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**A Basic Example Applying Machine Learning
to Sample Designs, Cont.**

- One possible solution:
 - Rather than take a simple random sample of X% of the records, it could be more efficient to first stratify the total VIP Customer frame.
 - But the auxiliary variables that are likely related to the key outcome of interest (time on social media mobile sources) are continuous.
 - We could create categorical versions of each of the three key predictors then cross them to form exhaustive strata
 - A possibly more efficient way could be to cluster the VIP customers using the three variables: total bill, total data consumed and mobile internet activity.
 - The resulting “clusters” become the strata for sampling.

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Big Data

Analyses of Big Data

Classical Statistical Approaches versus Statistical Machine Learning
Model Evaluation/Validation

Popular machine learning techniques

K-Means Clustering

K Nearest Neighbors

CARTS

Random Forests

Method Overview: K-nearest Neighbors

- **What type of Learner?**
 - ▢ **Nonparametric method; Supervised Learner (if y)**
 - ▢ **Unsupervised learner if no y or don't use it**
- **For what problems can you apply this method?**
 - ▢ **Regression/Prediction for Continuous Outcome Variables**
 - ▢ **Classification for Categorical Outcome Variables (2 or more levels)**
 - ▢ **Method usually applied when predictors are all continuous/ordinal**
- **Does this method have any Tuning Parameters?**
 - ▢ **One parameter: k – the number “neighbors”**
 - Here neighbors refer to cases in the data (i.e. rows in the D.S.)



Method Overview: KNN

● How does this Method Work?



Step 1: Determine Neighbors for Each Point in the Data Set

Using a distance measure (Euclidean/City), KNN identifies data points that are “close” in p-dimensional space to each case in the test set for which predicted values are desired.

Step 2: Make the Prediction/Estimate

For classification, the predicted value is the modal class of all k-nearest neighbors in the training set;

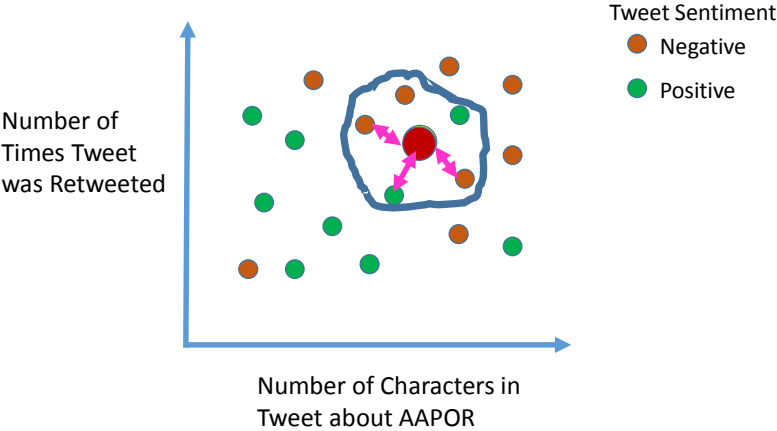
For regression, the predicted value is the mean/median of the outcome variable among all k-nearest neighbors in the training set.

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Example of KNN with 2 predictors and a Binary Outcome with K=5



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Method Overview: KNN

●

Insights on Parameters for this Method

▢

Small $k \rightarrow$ low bias, but high variance in estimates;

▢

Large $k \rightarrow$ higher bias but lower variance;

▢

For classification applications, k is usually chosen as an odd number (to avoid ties)

▢

Other distance functions can also be used, but most common is Euclidean Distance

●

Caveats/ Considerations for the Method

▢

This method is sensitive to scaling of p-predictors.

▢

If all predictors are on different scales/measured in different units, standardizing variables is strongly recommended.

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Method Overview: KNN

●

Caveats/ Considerations for the Method

▢

This method can be sensitive to the scaling of the p-predictors. If all are on different scales/ measured in different units, standardizing variables is recommended.

○

This effectively gives every predictor equal weight in determining “similarity” of cases.

▢

This method is not recommended for use with nominal categorical predictors without some additional preparation/transformation of these variables.

○

One alternative is k-mode clustering or providing similarity matrix for each case.

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Method Overview: KNN

● **Implementation of this method in R**

- train{caret package; use method="knn"};
- knn{class package};
- knn.reg{FNN package}



● **Examples of this method in the S.R. Literature**

- Multiple Imputation
 - Jonsson and Wohlin (2004); <http://www.wohlin.eu/metrics04.pdf>
- Image Classification
- Text Classification
- Propensity Weighting
- Binary Prediction from Longitudinal Survey Data
 - Bryant et al. (1996)
 - <http://irp.wisc.edu/publications/dps/pdfs/dp109296.pdf>

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Example 3: KNN for Classification

● **Imagine a mobile web survey just in time for spring.**

- The question asks respondents to choose their favorite type of flower from among a list of major types.
- Iris is one response option.
- Respondents are asked to capture a picture of their favorite Iris and upload the image via the survey.
- The goal is to process images to determine which *species* of Iris plant has been captured by the respondent (rather than asking them for the scientific species).



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Example 3: KNN, Cont.

- Let’s imagine 150 respondents answered Iris and provided photos of one of three common species of Iris plants: Setosa, Versicolor and Virginica.
- Using the images, we can generate measures of four continuous variables (all measured in cm):
 - ▣ Sepal Length; Sepal Width; Petal Length; Petal Width
 - ▣ Using these measurements we want to classify images into the three species groups.
 - ▣ Training a machine learning algorithm on respondent images will allow us to classify others in the future.
 - ▣ The specific dataset we will use here for this demonstration is the famous Fischer’s Iris data (>data(iris) in R)
http://en.wikipedia.org/wiki/Iris_flower_data_set

Some Beautiful Flowers...



Iris setosa

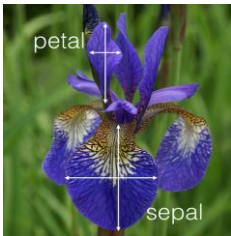


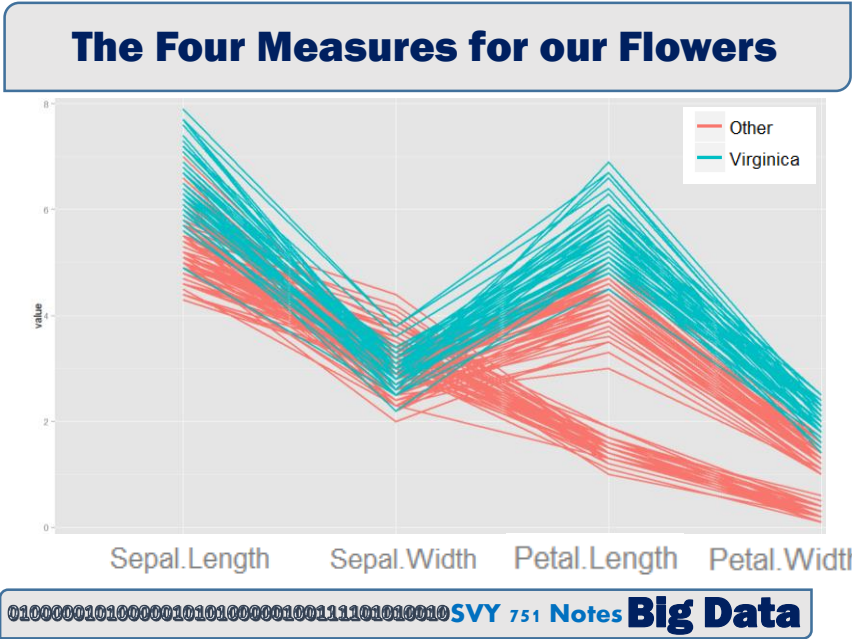
Iris versicolor



Iris virginica

GOAL: Develop a tool/train a machine learning method to classify flower images as Iris virginica or not.





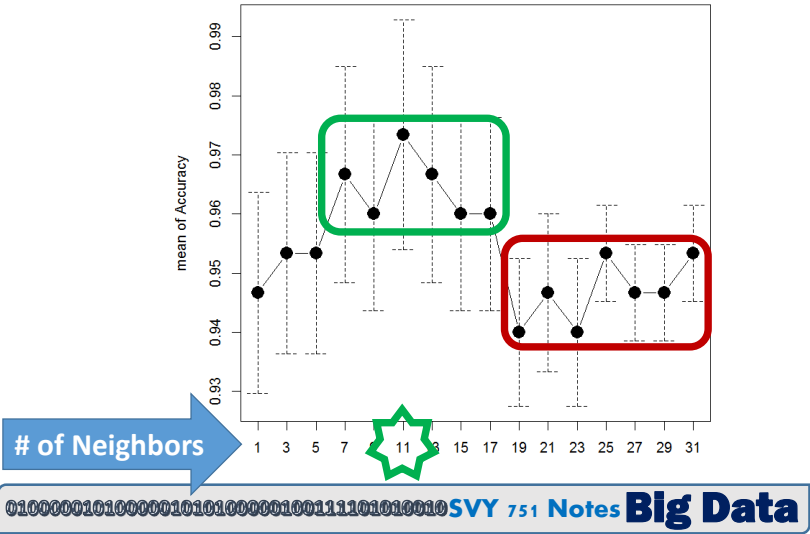
Run the KNN Classifier in R: Determine optimal number of Neighbors using 5-fold CV

```
# Run the k-nearest neighbor algorithm on iris data - 5-fold cross validation to #
estimate optimal k
set.seed(6152015)
irisctrl<-trainControl(method="cv", number=5, returnResamp="all")
irisknn<-train(iris[,c(1:4)],y=iris$Species2, method="knn",
tuneGrid=data.frame(.k=c(2*(0:15)+1)),trControl=irisctrl)
# Plot the resulting values of accuracy versus k - cross validated averages
# Plot not shown, but matches the mean values on plot on next slide
plot(irisknn, lwd=3, col=2)
# Means plots representing 5-fold averages/standard deviations for accuracy
iriscvresults<-irisknn$resample
iriscvresults<-iriscvresults[order(iriscvresults[,3], iriscvresults[,4]),]
# FIGURE ON NEXT SLIDE
iriscvresults <- within(iriscvresults, {kfact <- as.factor(k)})
with(iriscvresults, plotMeans(Accuracy, kfact, error.bars="se", col=2, lwd=3))
# Determine the CV-Optimal Neighborhood size:
irisknn$bestTune # 11 - optimal # of neighbors for classification based on 5-fold CV
```

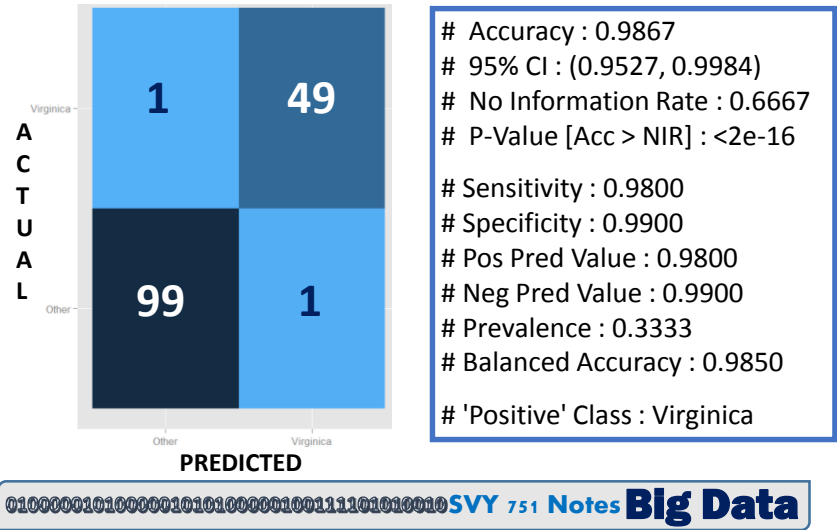


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5-Fold Cross Validation to Determine the Optimal Number of Neighbors



Overall Results of the KNN Model for Iris Data



Analyses of Big Data

- Classical Statistical Approaches versus Statistical Machine Learning
- Model Evaluation/Validation
- Popular machine learning techniques
 - K-Means Clustering
 - K Nearest Neighbors
 - CARTS
 - Random Forests

Method Overview: Classification and Regression Trees (CARTs)

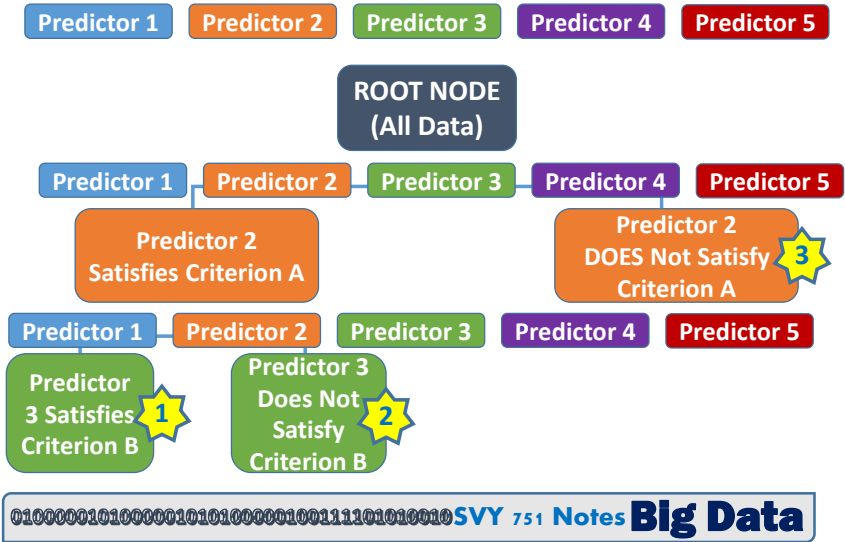
- **What type of Learner?**
 - ▢ Nonparametric method; Supervised Learner (if y)
 - ▢ Unsupervised learner if no y or don't use it
 - ▢ CARTs a.k.a. *RECURSIVE Partitioning*
- **For what problems can you apply this method?**
 - ▢ Regression/Prediction for Continuous Outcome Variables
 - ▢ Classification for Categorical Outcome Variables (2 or more levels)
 - ▢ Tree methods can handle predictors that are continuous, categorical, skewed and sparse data as well as missing data
- **Does this method have any Parameters?**
 - ▢ Complexity Parameter (Cp)
 - ▢ Node Size/Branches/Fraction of Objects



How Do CARTs Work?

- Step 1: Root Node Split.** Starting with the entire data set (root node), select a variable (and corresponding range or grouping of levels) that generates two subsequent nodes that are each more “pure” than the root node.
- The usual optimization criterion for tree growing is Node Impurity is measured using the SSR (sum of squared residuals/errors) for continuous outcomes and using the Gini Index or Chi Squared/Deviance criterion for categorical outcomes.
- Step 2: Continued Branching.** Within each of the subsequent nodes, additional splits (branching) are determined from all possible predictors; any resulting child nodes are even more homogenous/pure than parent nodes.
- Step 3: Pruning.** After a tree is grown to completion, pruning branches back according to the complexity parameter is sometimes applied. Essentially, nodes and branches are removed from the tree producing a tree that is not as deep and with fewer final nodes.
- Step 4: Prediction.** Within each final node the predicted value is either the modal value/class of the outcome (Classification) or the mean of the outcome variable for observations in the node (Regression)

CART Overview



Method Overview: CARTs

●

Insights on Parameters for this Method

Small C_p → lead to more complex trees (larger, deeper with more final nodes/leaves).

Large C_p → lead to less complex trees (smaller, more shallow with fewer final nodes/leaves)

●

Caveats/ Considerations for the Method

Trees that are too large or overly complex can result in estimates that perform poorly for new data (low bias but high variance).

Conversely, trees that are not complex enough may result in estimates that don't adequately fit (lower variance but possibly larger bias).

Missing data are ignored when the tree is being built. However, when used for prediction, tree models can handle missing data by using SURROGATE variables – those related to the splitting variable and for which results are similar.

Predictors with large numbers of categories tend to get selected for splitting compared to those with fewer categories.

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Method Overview: CARTs

●

Advantages of this Method

Tree-based models are nonparametric and don't assume a functional form between outcome and predictors

Tree-based models naturally detect higher level interactions among the predictors.

Tree-based models are easy to visualize and interpret – even if the number of candidate predictors is large

●

Disadvantages of this Method

Single trees are likely to have sub-optimal predictive performance compared to other methods (Kuhn and Johnson, 2013)

- This is especially true if the relationship between the outcome and predictors is not described well by "rectangular" boundaries

The number of different predicted values (for regression trees) is determined by the number of final nodes in the tree.

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Method Overview: CARTs

- Implementation of this method in R
 - ▢ `rpart`{`rpart` package}
 - ▢ `tree`{`tree` package};
 - ▢ `ctree`{`party` package}
 - ▢ `Rattle` {`rattle` package}
- Examples of this method in the S.R. Literature
 - ▢ **Response Propensity Estimation and Nonresponse Bias Evaluation**
 - Phipps and Toth, 2012; <http://1.usa.gov/1l4fObz>
 - McCarthy et al. (2009); <http://1.usa.gov/1lcQekK>
 - Schouten and de Nooij (2005); <http://bit.ly/1zUrPtf>
 - ▢ **Frame Revision/Trimming**
 - McCarthy et al. (2009); <http://1.usa.gov/1lcQekK>
 - ▢ **Multiple Imputation**
 - Mesa, Tsai and Chambers (2000); <http://bit.ly/1DrkSBt>
 - ▢ **Generating Partially Synthetic Data**
 - Reiter (2005); <http://bit.ly/1DY6kvz>

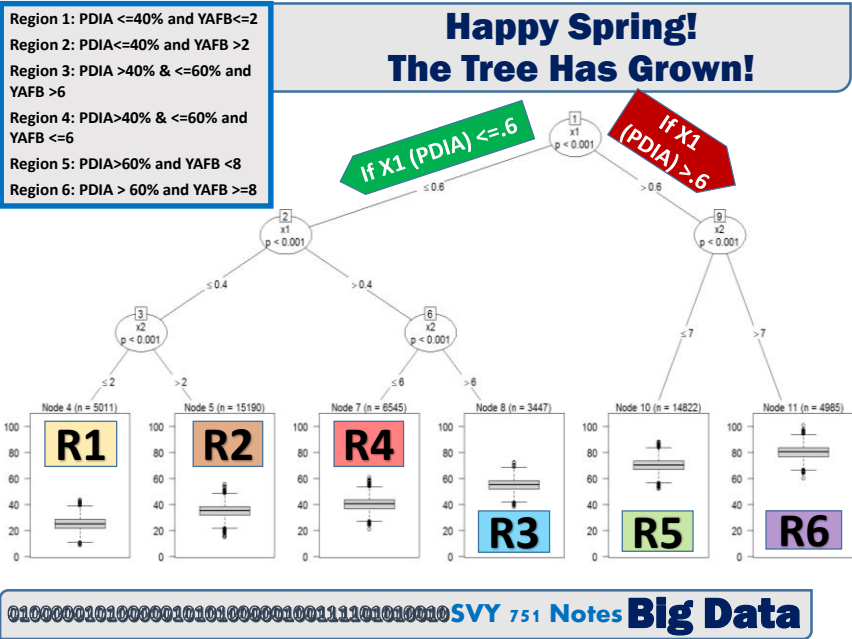
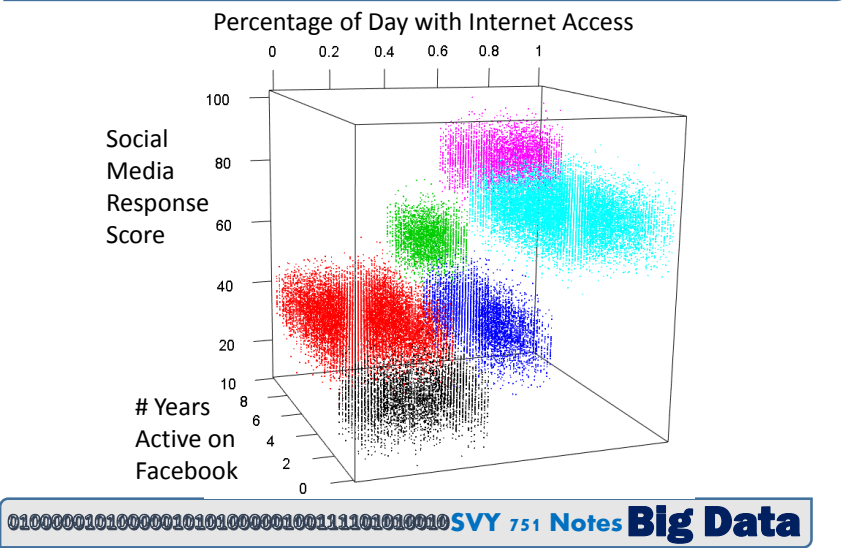
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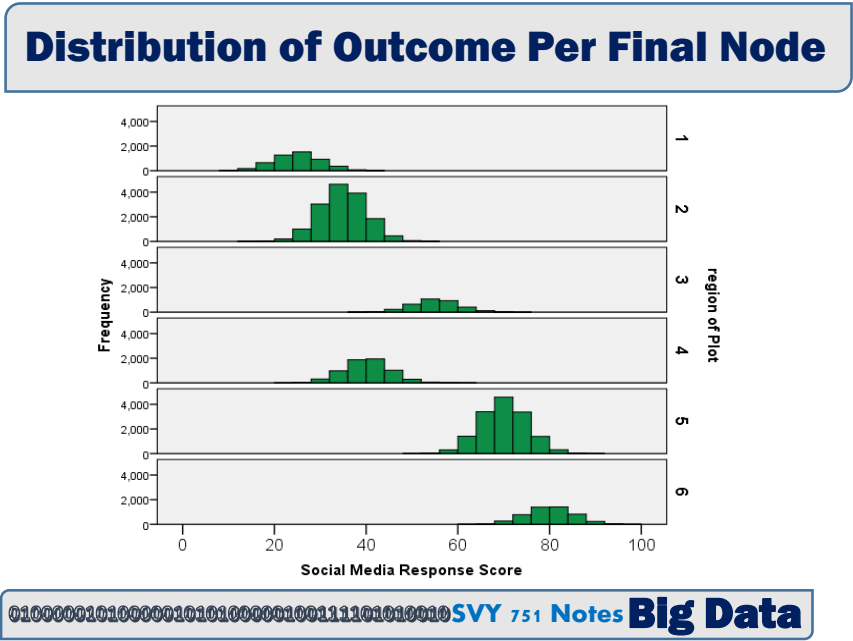
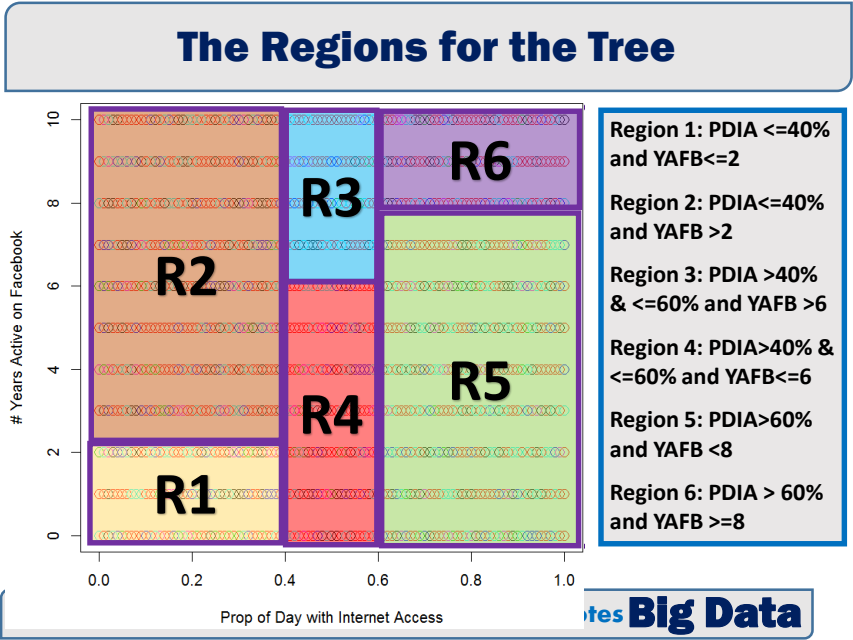
Example 4: Regression Trees

- A media company specializing in online advertisement and survey recruiting via social media wants to estimate a new metric called *Social Media Response Score*.
 - ▢ This metric measures how likely a user is to respond to promptings, invitations and solicitations via social media advertising methods.
 - ▢ Two predictors of this metric being evaluated include:
 - The Number of Years active on Facebook (0 – 10);
 - The proportion of the day the user has access to internet (0-1).

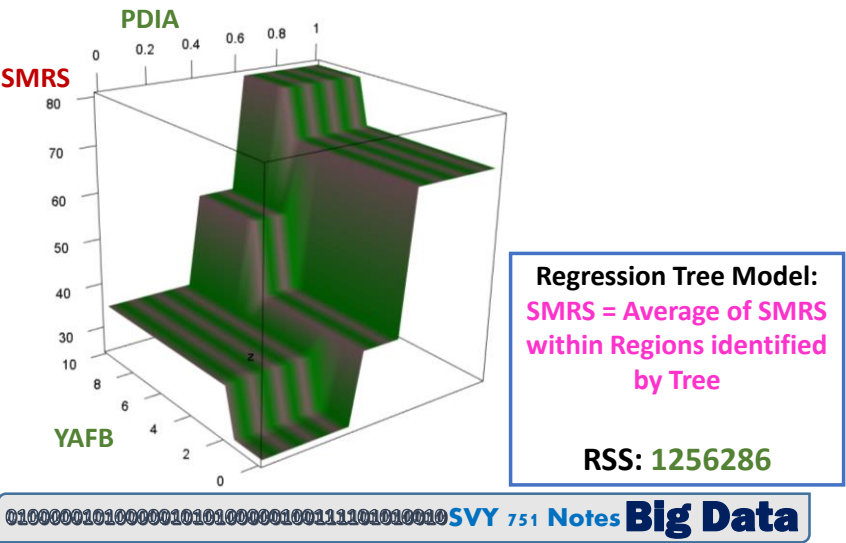
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The Overall Data Set (n=50,000)

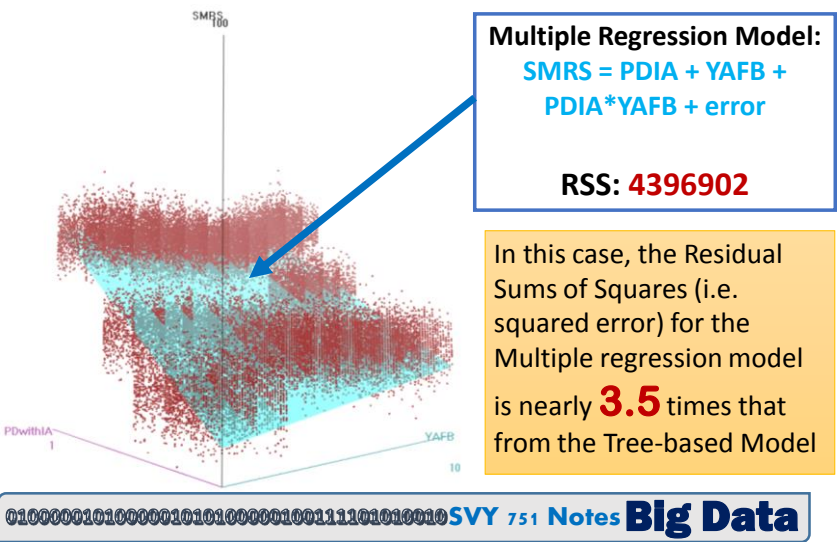




Tree-Based Model Results



Multiple Regression Based Results



R-code for Example 4: Regression Trees

```
# TreeExample Data: x1=PDIA; x2=YAFB and y=Social Media Response Score
# region corresponds to one of the 6 regions defined by the specific combinations of x1 and x2
#Plot the surface defined by the three variables x1, x2 and y.
require(rgl)
plot3d(TreeExample$x1, TreeExample$x2, TreeExample$y, col=TreeExample$region, size=2, xlab=" ", ylab=" ", zlab=" ")

# There are many options for creating trees, we choose ctree here from the party package
# primarily because of the default features of it's resulting graphics
require(party)
set.seed(52015)
tree.Example<-ctree(y~x1+x2, data=TreeExample)
# Now evaluate the Residual Sums of Squares from the Predictions
RSSmodel<-sum((TreeExample$y-predict(tree.Example))^2)
[1] 1256286

# This code allows us to plot the "predicted" surface estimated from the tree
f <- function(x, y) {newdat<-data.frame(x, y)
names(newdat)<-c("x1", "x2")}
fout<-predict(tree.Example, newdata=newdat)
fout}

x1dat<-seq(0,1, length.out=20)
x2dat<-seq(0,10, length.out=20)
z<-outer(x1dat, x2dat, f)
persp3d(x=seq(0,1, length.out=nrow(z)), y=seq(0,10, length.out=ncol(z)),z, col=c("pink", "green2"), xlab=" ", ylab=" ")
```



Some Beautiful Flowers...REDUX



Iris setosa

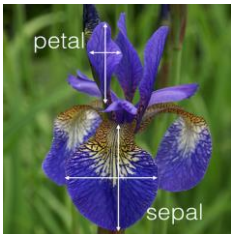


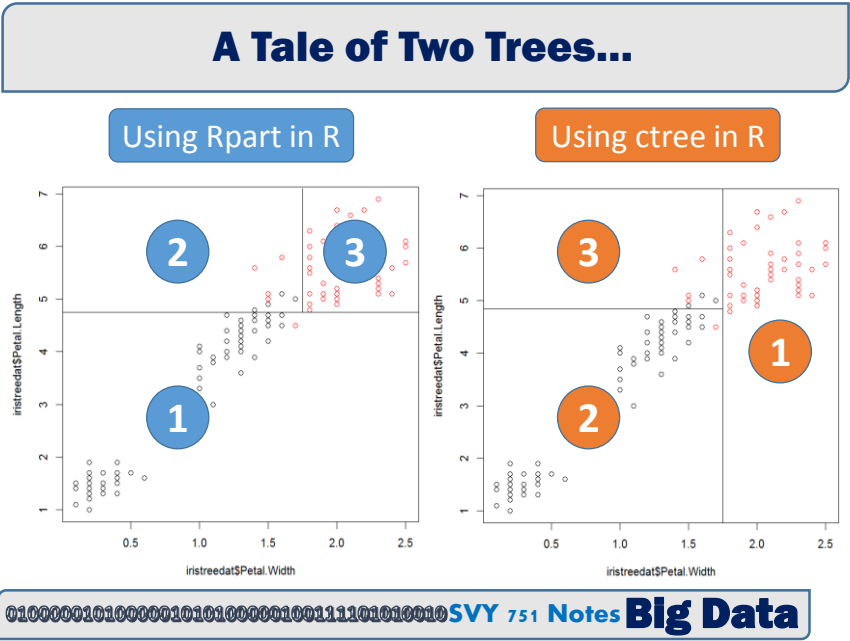
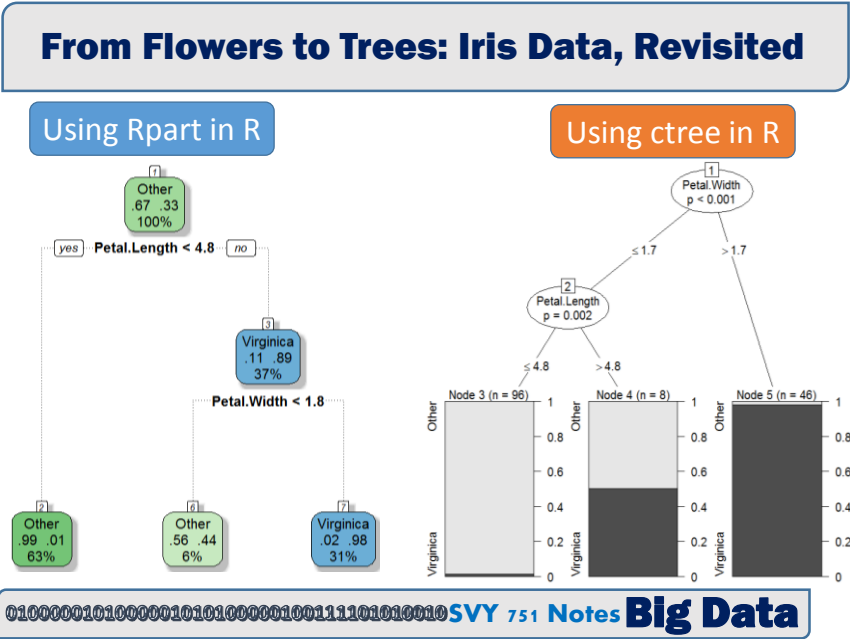
Iris versicolor



Iris virginica

GOAL: Develop a tool/train a machine learning method to classify flower images as Iris virginica or not.





Overall Results of the Tree Model for Iris Data

TREE Model

Accuracy : 0.96
95% CI : (0.915, 0.9852)
No Information Rate : 0.6667
P-Value [Acc > NIR] : <2e-16

Sensitivity : 0.9000
Specificity : 0.9900
Pos Pred Value : 0.9783
Neg Pred Value : 0.9519

'Positive' Class : Virginica

KNN (11) Model

Accuracy : 0.9867
95% CI : (0.9527, 0.9984)
No Information Rate : 0.6667
P-Value [Acc > NIR] : <2e-16

Sensitivity : 0.9800
Specificity : 0.9900
Pos Pred Value : 0.9800
Neg Pred Value : 0.9900

'Positive' Class : Virginica

R-code for Generating Tree Model for Iris Data

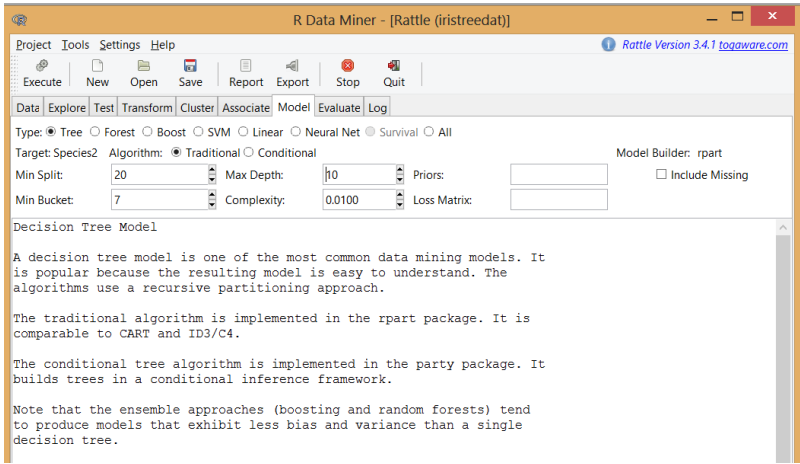
```
require(rpart)
# create simple iris data set with only one binary outcome
# -5 removes 5th column from iris which is the three-level outcome we are not using here.
iristreedat<-iris[,-5]
require(rpart) # rpart is one package for generating trees
set.seed(510)
iristree<-rpart(formula = Species2 ~ ., data = iristreedat, method = "class", parms = list(split =
"information"), control = rpart.control(maxdepth = 10, usesurrogate = 0, maxsurrogate = 0))
#Plot the tree (ON NEXT SLIDE)
fancyRpartPlot(iristree, main="Decision Tree iristreedat $ Species2")

# Now apply the tree to generate the predicted classes for each
iristreepred<-predict(iristree, newdata=iristreedat[,c(1:4)], type="class")
iristreeconfuse<-confusionMatrix(iristreepred, iristreedat$Species2, positive=c("Virginica"))

# Alternative tree [via Party Package] (conditional tree that does not require a cp parameter)
set.seed(510)
iristree2<-ctree(Species2 ~ ., data = iristreedat)
plot(iristree2)
```



Machine Learning in R via Rattle!



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Analyses of Big Data

- Classical Statistical Approaches versus Statistical Machine Learning
- Model Evaluation/Validation
- Popular machine learning techniques
 - K-Means Clustering
 - K Nearest Neighbors
 - CARTS
 - Random Forests

Method Overview: Random Forests

●

What type of Learner?

▢

Nonparametric method; Supervised Learner (if y)

▢

Unsupervised learner if no y or don't use it

▢

Random Forests fall under a class of learners called "Ensemble Methods"

●

For what problems can you apply this method?

▢

Regression/Prediction for Continuous Outcome Variables

▢

Classification for Categorical Outcome Variables (2 or more levels)

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Method Overview: Random Forests

●

Does this method have any Parameters?

▢

Number of Trees (ntree)

▢

Variables to be randomly considered at each node (mtry)

▢

Node sizes

●

Insights on Parameters for this Method

▢

Node size is typically 1 for classification and 5 (or more) for regression.

▢

Generally, the more trees in the forest, the less error one has. But after a certain point, diminishing returns on error rate improvements are likely to be found in practice.

▢

Larger values of mtry increases the strength of the tree, but also the correlation between the results of one tree from another in the forest.

▢

Smaller values of mtry reduces the inter-tree correlation as well as the overall predictive strength of the tree.

○

Default mtry is set as $p/3$ for regression

○

Default mtry is set as $\sqrt{p}$ (rounded down) for classification

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How Do Random Forests Work?

Step 1: Establish the Forest. Generate n_{tree} bootstrap samples from the original dataset with replacement and of the same size as the input dataset.

Step 2: Grow Trees. For each of the bootstrap samples drawn in Step 1, grow a classification/regression tree to full size (i.e. no pruning). But at **each** node of each of these trees, randomly select m_{try} of the predictor variables to be used as the basis of branching. Splits/branches are based on the best variable from among the m_{try} that are selected for a given node.

Step 3: Prediction. After all trees in the forest have been grown, form a prediction for new data from each tree in the same way as you would for normal classification or regression trees.

For forests based on regression trees, final predicted values for new data are computed as the mean of these n_{tree} predicted values;

For forests based on classification trees, final predicted values for new data are computed as the class with the majority vote across the n_{trees} (where each tree in the forest contributes one vote for a single outcome class).

Random Forests Illustrated...

Tree 1: YES Tree 2: YES Tree 3: NO

Forest Uses Majority Rule: Prediction for New Data: YES

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Error Rate Estimation for Random Forests

●How are error rates assessed for Random Forests?

- ▢ Unlike other machine learning methods that use some type of external test sample to assess performance, Random Forests generate an internally valid, and in many cases unbiased, assessment of error.
- ▢ The error rate is typically referred to as the Out of Bag Error Rate (OOB Error Rate).

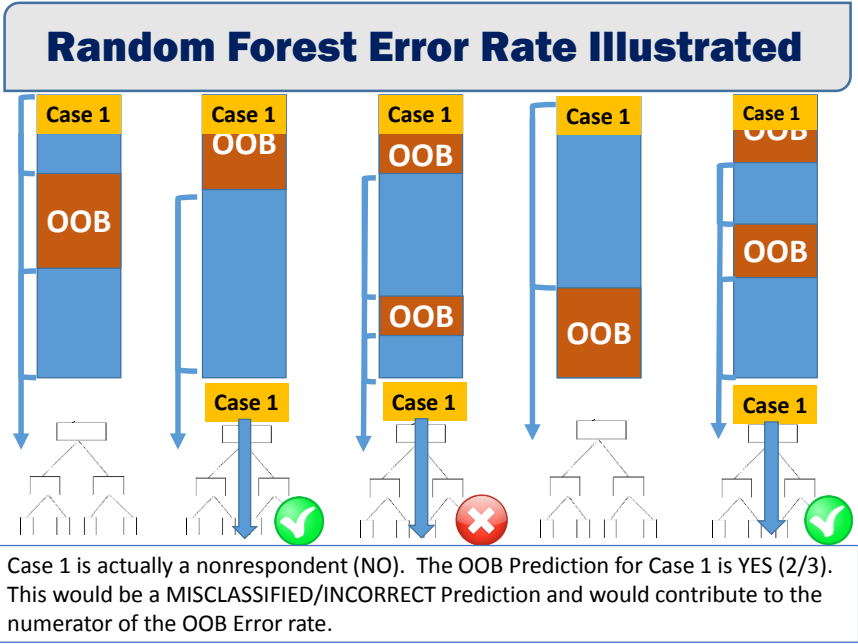


Error Rate Estimation for Random Forests

●How is the OOB Error Rate Computed?

- ▢ Step 1: Each tree in the forest uses approximately 2/3 of the original data. For the remaining 1/3 of cases, (called Out of Bag, OOB), use the given tree to make predictions. Call these “Out of Bag Predictions.”
- ▢ Step 2: For a given case, aggregate all “Out of Bag Predictions” across the roughly $n_{tree}/3$ trees for which the case was OOB.
 - Averages for regression applications
 - Majority vote for classification applications.
- ▢ Step 3: Compare the aggregated estimate from Step 2 to the actual value of the outcome for each case in the data set
 - Correct/Incorrect for Classification
 - Residual for Regression ($y_i - \text{Step 2 Estimate}$).
- ▢ The OOB Error is the:
 - proportion of “incorrect” values from Step 3 (classification)
 - the average squared residuals from Step 3 (regression)





Method Overview: Random Forests

Caveats/ Considerations for the Method

- For situations where the number of predictors is greater than the number of cases ($n < p$), mtry values should be increased beyond the defaults.
 - See: https://samos.univ-paris1.fr/IMG/pdf_S1E1-Poggi.pdf
- Random Forests do not produce coefficients or significance tests for predictors. They do produce variable importance measures that can help rank order the utility of the variables in predicting the outcome.
- Random Forests don't use surrogates like trees, so missing data must be dealt with
- Variable Importance measures can be biased if the predictor variables are either highly correlated or on different scales.
 - Suggest using Conditional Forests instead (Strobl et al., 2007)
 - See: <http://www.biomedcentral.com/1471-2105/8/25>

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Method Overview: Random Forests

- **Advantages of this Method**
 - ▢ Like Tree methods, Random Forests can handle predictors that are continuous, categorical, skewed and sparse data as well as missing data.
 - ▢ Random forests are also aptly suited for the “large p, small n” scenario (Strobl et al., 2007).
 - ▢ Random forests can also be very effective for estimating outcomes that are a complex functions of predictors with many interactions or possibly non-linear functions of the parameters (Mendez et al., 2008).
- **Disadvantages of this Method**
 - ▢ Random Forests can be computationally intensive.
 - ▢ Unlike Trees, Random Forests are not easily visualized.



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Method Overview: Random Forests

- **Implementation of this method in R**
 - ▢ `randomforest{randomForest package}`
 - ▢ `rfsrc{randomForestSRC + ggRandomForest packages}`
 - ▢ `cforest{party package}`
 - ▢ `CoreModel{CORElearn package}`
- **Examples of this method in the S.R. Literature**
 - ▢ **Response Propensity Weighting and Nonresponse Bias Evaluation**
 - Buskirk and Kolenikov (2015); <http://surveyinsights.org/?p=5108>
 - Buskirk, West and Burks (2013); <http://bit.ly/1JVzgWt>
 - ▢ **Hybird Ensemble Method for Response Propensity Scores and Nonresponse Bias evaluation for Establishment Surveys**
 - Earp, Mitchell, McCarthy and Kreuter (2014); <http://bit.ly/1Pi4Glf>
 - ▢ **Identifying important predictors of county-level CPO rates**
 - Buskirk, et al. (2014); <http://bit.ly/1Jd3TXt>



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Buskirk and Kreuter, SYV 751

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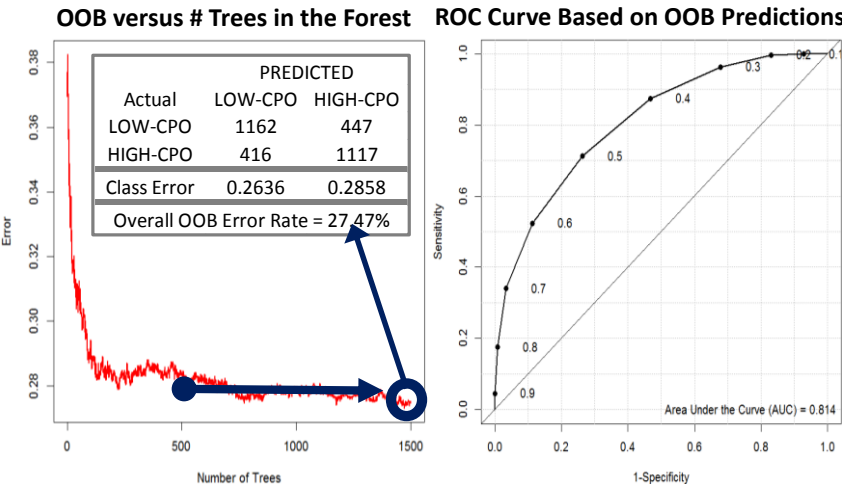
Example 5: Random Forests for Classification

- **MSG Estimated County-Level CPO Rates for 3,142 Counties 2013 Q1**
 - classified as either CPO-HIGH/CPO-LOW based on whether the estimated county-level CPO rate was below the national average estimate (31.87%)
- **Demographic counts:**
 - Race, Sex, Ethnicity, Age, %HH with Children, Education, etc.
- **Economic Indicators**
 - Annual Median Income, # Vehicles/Household, Families below Poverty
- **Housing/Area Attributes**
 - Counts of Dwelling Units, Group Quarters, Families/Non-Families; Land Size
- **Proportion of residential addresses in the county in:**
 - Urban Clusters, Suburban Rings or Outlying Rural areas.
- **Spatial Correlation Index 2013 (Q1) CPO rates (Moran's I)**
- **Census' Low Response Indicator for the County**
- **Combinations of variables within and across these groups**
 - % AA HHs w/ Female Householder with Children, Single Dwelling Unit
- **More information is contained in Buskirk, et al. (2014); <http://bit.ly/1Jd3TXt>**

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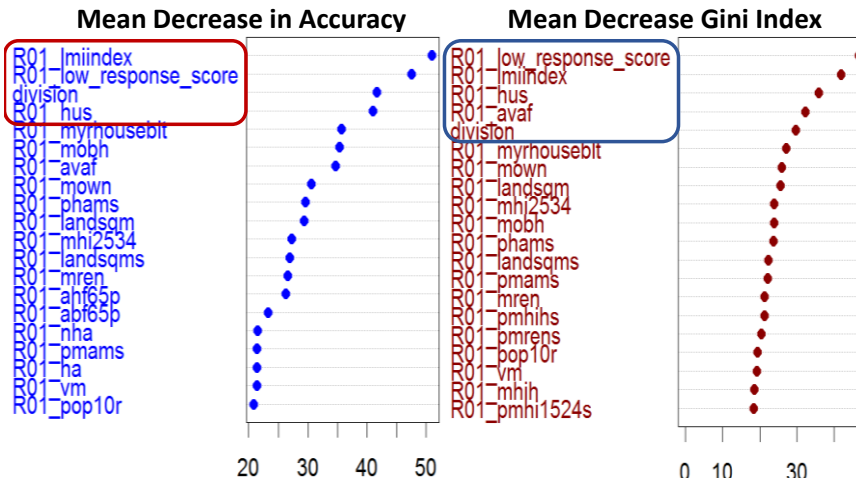
OOB Performance Results for Random Forest Example



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Variable Importance for Random Forest Example



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R-code for Random Forest Example

```
require(randomForest, quietly=TRUE)
# Build the Random Forest model: There are 56 predictors, so mtry defaults to floor(sqrt(56))=7.
set.seed(42)
rfexample<- randomForest(cpohigh~., data=cpo2013[,c(2:58)], ntree=1500, mtry=7,
  importance=TRUE, na.action=na.omit, replace=FALSE)

#Plot the Out-of-Bag Error rate as a function of the number of trees in the forest
plot(x=c(1:1500),y=rfexample$err.rate[,1], xlab="Number of Trees", ylab="Error", type="n")
lines(x=c(1:1500),y=rfexample$err.rate[,1], xlab="Number of Trees", ylab="Error", col=2, lty=1, lwd=2)
# Determine the variable importance measures: these were stored in rfexample with importance=TRUE.
rfimport<-round(importance(rfexample), 2)

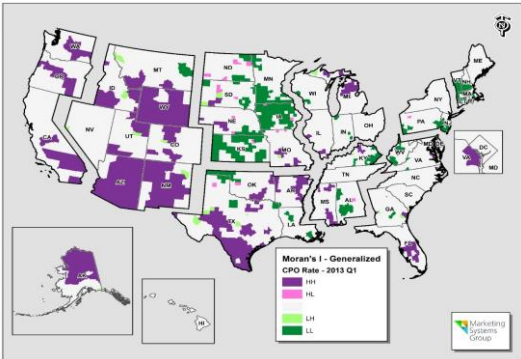
#Plot the variable importance measures - top 30 variables
# First Plot is mean decrease in accuracy (based on OOB Permutations)
varImpPlot(rfexample, type=1, col=4, sort=TRUE, n.var=20, cex=1.5, pch=16)
# Second plot is based on avg. decrease in the Gini Index
varImpPlot(rfexample, type=2, col="darkred", sort=TRUE, n.var=20, cex=1.5, pch=16)
title(main="Variable Importance Random Forest cpo2013", cex=2)

# Plot the ROC curve corresponding to the Random Forest
require(pROC, quietly=TRUE)
aucc<-roc(as.numeric(rfexample$y)-1, rfexample$votes[,2], thresholds=seq(from=0, to=1, by=.1))
roc.plot(x=as.numeric(rfexample$y)-1, pred=rfexample$votes[,2], thresholds=seq(from=0, to=1, by=.1),
  ylab="Sensitivity", xlab="1-Specificity")
legend("bottomright", bty="n",sprintf("Area Under the Curve (AUC) = %1.3f", aucc$auc))
```



Random Forest Example, Cont.

- The Moran's I values can be used to generate geospatial clusters including HH, HL, LH and LL
 - HH/LL indicates areas in which the county CPO rates are high/low and surrounded by other counties with high/low CPO rates, respectively.

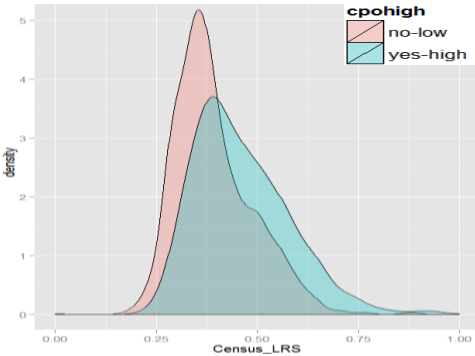


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Scaled Census Low Response Score by CPO High/Low Status

Summary Statistic	CPO Status	
	no-low	yes-high
mean	18.37	20.17
sd	2.53	3.09
IQR	3.12	4.19
Median	17.84	19.66
Min	8.17	13.60
Max	31.33	34.19
n	1578	1564



```
# Descriptive Statistics on Rescaled Census Response Score by CPO Status
numSummary(cpo2013[, "R01_low_response_score"], groups=cpo2013$cpohigh,
statistics=c("mean", "sd", "IQR", "quantiles"), quantiles=c(.5,0,1))
#Look at the Scaled Census Low Response Score by CPO Status
cposhort<-data.frame(cpo2013$cpohigh, cpo2013$R01_low_response_score)
names(cposhort)<-c("cpohigh", "Census_LRS")
ggplot(data=cposhort, aes(x=Census_LRS, group=cpohigh, fill=cpohigh)) + geom_density(alpha=.3)
```



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Appendix 2: Variable Description for Final 56 Predictors for Random Forest Example

Note 1: Each of the predictors were rescaled to be on the same scale – [0,1] by subtracting the minimum value of the predictor from each case and dividing this difference by the range of the predictor. In the R dataset cpo2013, the rescaled variables names have the prefix “R01_” to indicate they have been rescaled.

Note 2: Variables ending in “M” represent the county mean of the variable computed as the average value of the variable across all the block groups in the county. If the variable name ends in “S” the predictor represents the Standard Deviation of the specified variable computed as the standard deviation of the variable across all block groups in the county.

Predictor	Variable Name	Label
1	LMIIIndex	Moran's I Geospatial Similarity Measure
2	division	Census Division
3	MYRHouseBlt	Current Year Median Year, Housing Unit Structure Built
4	Low_Response_Score	Census Low Response Score
5	HUS	Current Year Housing Units STD
6	AVAF	Current Year Average Age, Female
7	MOWN	Current Year Median Householder Years of Residence, Owner-Occupied Housing Units
8	MHI2534	Current Year Median Household Income, Householder Age 25 - 34
9	MOBH	Current Year Housing Units, Mobile Home
10	LANDSQM	Land Area in Square Miles
11	PMHIHS	Current Year Median Household Income, Hispanic/Latino Householder STD
12	HA	Current Year Average Household Size
13	NHA	Current Year Average Household Size, NonFamily Household
14	PHAMS	Current Year Median Age, Householder STD
15	LANDSQMS	Land Area in Square Miles STD

Appendix 2 cont. Variable Description for Final 56
Predictors for Random Forest Example

Predictor	Variable Name	Label
16	PMAMS	Current Year Median Age, Male STD
17	MREN	Current Year Median Householder Years of Residence, Renter-Occupied Housing Units
18	POPS	Current Year Population STD
19	AHF65P	Current Year Population, Hispanic/Latino, Female: Age 65+
20	PMHI85PS	Current Year Median Household Income, Householder Age 85+ STD
21	ABF65P	Current Year Population, Black/African American Alone, Female: Age 65+
22	PMHI1524S	Current Year Median Household Income, Householder Age 15 - 24 STD
23	VM	Current Year Median Value, Owner-Occupied Housing Units
24	POP10R	Census 2010 Population : Rural (Non-Urban)
25	MHI1524	Current Year Median Household Income, Householder Age 15 - 24
26	PGQOTHs	Current Year Group Quarters Population, Institutional: Other Facilities STD
27	PMRENS	Current Year Median Householder Years of Residence, Renter-Occupied Housing Units STD
28	PMRESS	Current Year Median Householder Years of Residence STD
29	MHI85P	Current Year Median Household Income, Householder Age 85+
30	GQPOPS	Current Year Group Quarters Population STD
31	PCT_U1	City Center, Urban
32	UOTH	Current Year Housing Units, Other Unit Type
33	GQOTH	Current Year Group Quarters Population, Institutional: Other Facilities
34	PMHI7584S	Current Year Median Household Income, Householder Age 75 - 84 STD
35	PCT_S2	Mid Suburbs
36	PMHINHs	Current Year Median Household Income, Not Hispanic/Latino Householder STD
37	PV2040S	Current Year Owner-Occupied Housing Units, Value \$20,000 - \$39,999 STD

Appendix 2 cont. Variable Description for Final 56
Predictors for Random Forest Example

Predictor	Variable Name	Label
38	PVMS	Current Year Median Value, Owner-Occupied Housing Units STD
39	PCT_R	ALL Rural
40	V4060	Current Year Owner-Occupied Housing Units, Value \$40,000 - \$59,999
41	V1MP	Current Year Owner-Occupied Housing Units, Value \$1,000,000+
42	MHIH	Current Year Median Household Income, Hispanic/Latino Householder
43	U1A	Current Year Housing Units, 1 Unit Attached
44	PMYRBS	Current Year Median Year, Housing Unit Structure Built STD
45	GQCOL	Current Year Group Quarters Population, Noninstitutional: College/University Student Housing
46	PMHI2534S	Current Year Median Household Income, Householder Age 25 - 34 STD
47	U2	Current Year Housing Units, 2 Units
48	AAF65P	Current Year Population, Asian Alone, Female: Age 65+
49	PCT_U2	ALL Other, Urban
50	PCT_S1	Inner Suburbs
51	POP10UC	Census 2010 Population : Urban : Urban Cluster
52	WTPT	Current Year Workers, Transportation To Work: Public Transport
53	WTB	Current Year Workers, Transportation To Work: Bicycle
54	FAF	Current Year Population 16+, Female: Armed Forces
55	PCT_S3	Outer Suburbs
56	NFFC	Current Year Households, w/ People < 18 Years Old, NonFamily Household, Female Householder

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